

- RAG Learning

2.2 Formulae, Identities, Proof & Functions

Pearson Edexcel GCSE Mathematics 1MA1 — Algebra — spec refs A5–A7 — Papers 1 (non-calculator), 2 and 3 (calculator)

2.2.1 Using Standard Formulae (A5)

A **formula** connects two or more variables. You will meet formulae written in symbols and described in words, from maths, science and everyday contexts. Some formulae have to be learned (area of a trapezium, volume of a prism, Pythagoras); others are printed in the question or on the formulae page.

Subject: the variable written on its own on one side of a formula. In $F = ma$, F is the subject.

Substitution: replacing each letter with its value, then evaluating in the correct order of operations.

Kinematics formulae (given in the exam when needed), where u = initial velocity, v = final velocity, a = constant acceleration, t = time taken, s = displacement:

$$v = u + at \quad s = ut + \frac{1}{2}at^2 \quad v^2 = u^2 + 2as$$

Worked Example: Use $s = ut + \frac{1}{2}at^2$ to work out s when $u = 4$, $t = 5$ and $a = 1.2$.

$$ut = 4 \times 5 = 20$$

$$\frac{1}{2}at^2 = \frac{1}{2} \times 1.2 \times 25 = 15 \text{ (square } t \text{ before multiplying)}$$

$$s = 20 + 15 = 35 \text{ m}$$

Worked Example: A taxi charges £3 plus £1.50 for every mile. Write a formula for the cost, £ C , of a journey of m miles and work out the cost of an 8-mile journey.

$$C = 3 + 1.5m$$

$$C = 3 + 1.5 \times 8 = 3 + 12 = 15, \text{ so the journey costs £15}$$

Worked Example: Use $v^2 = u^2 + 2as$ to work out v when $u = 6$, $a = 2$ and $s = 7$, given $v > 0$.

$$v^2 = 6^2 + 2 \times 2 \times 7 = 36 + 28 = 64$$

$$v = \sqrt{64} = 8$$

Exam Tip: Write down the formula with the numbers substituted before you press any calculator keys. If the final answer is wrong, the substitution can still earn M1.

2.2.2 Changing the Subject of a Formula (A5)

Rearranging uses the same balancing method as solving an equation: apply the **inverse operation to both sides**. Undo the operations on the new subject in the reverse of the order they were applied.

Done to the subject	Undo by...
adding k	subtracting k
multiplying by k	dividing by k
dividing by k	multiplying by k

squaring	square-rooting ($\pm$ unless the variable is known to be positive)
square-rooting	squaring
cubing	cube-rooting

Worked Example: Make a the subject of $v = u + at$.

Subtract u : $v - u = at$

Divide by t : $a = \frac{v - u}{t}$

Worked Example: Make x the subject of $y = \frac{2x + 7}{3} - 4$.

Add 4: $y + 4 = \frac{2x + 7}{3}$

Multiply by 3: $3y + 12 = 2x + 7$

Subtract 7: $3y + 5 = 2x$

Divide by 2: $x = \frac{3y + 5}{2}$

Powers and roots

Worked Example: The surface area of a sphere is $A = 4\pi r^2$. Make r the subject, where $r > 0$.

Divide by 4π : $r^2 = \frac{A}{4\pi}$

Square root: $r = \sqrt{\frac{A}{4\pi}}$

Worked Example: Make v the subject of $E = \frac{1}{2}mv^2$, where $v > 0$.

Multiply by 2: $2E = mv^2$

Divide by m : $v^2 = \frac{2E}{m}$

Square root: $v = \sqrt{\frac{2E}{m}}$

Exam Tip: Get the squared term on its own *before* taking the square root, and keep the whole of the other side under the root sign. $\sqrt{a+b}$ is not $\sqrt{a} + \sqrt{b}$.

2.2.3 When the Subject Appears More Than Once (A5) Higher

If the new subject appears in two terms, gather those terms together and factorise:

1. Remove fractions by multiplying both sides by the denominator; expand any brackets.
2. Move every term containing the subject to one side and all other terms to the other side.
3. Factorise out the subject.
4. Divide by the bracket.

Worked Example: Make x the subject of $px + 5 = qx - 2$.

Collect x terms: $px - qx = -2 - 5 = -7$

Factorise: $x(p - q) = -7$

Divide: $x = \frac{-7}{p - q} = \frac{7}{q - p}$

Worked Example: Make x the subject of $y = \frac{2x - 1}{x + 3}$.

Multiply by $(x + 3)$: $y(x + 3) = 2x - 1$

Expand: $xy + 3y = 2x - 1$

Collect x terms: $xy - 2x = -1 - 3y$

Factorise: $x(y - 2) = -1 - 3y$

Divide (and tidy the signs): $x = \frac{1 + 3y}{2 - y}$

Exam Tip: Test your answer with a number. If $x = 2$, then $y = \frac{3}{5} = 0.6$, and

$$\frac{1 + 3(0.6)}{2 - 0.6} = \frac{2.8}{1.4} = 2 \checkmark.$$

2.2.4 Equations and Identities (A6)

Equation: true only for particular values. $5x - 2 = 13$ is true only when $x = 3$.

Identity: true for every value of the variable, written with $\equiv$ ("is identically equal to").

Example: $4(x - 2) \equiv 4x - 8$.

Showing that two expressions are equivalent

Start from one side (usually the more complicated one) and simplify it step by step until it matches the other side. Every line of working must be shown; jumping straight to the answer earns nothing on a "show that" question.

Worked Example: Show that $(x + 6)^2 - (x - 6)^2 \equiv 24x$. Hence work out $106^2 - 94^2$ without a calculator.

$$(x + 6)^2 = x^2 + 12x + 36 \text{ and } (x - 6)^2 = x^2 - 12x + 36$$

$$x^2 + 12x + 36 - x^2 + 12x - 36 = 24x \checkmark$$

$$\text{With } x = 100: 106^2 - 94^2 = 24 \times 100 = 2400$$

Finding unknown constants

Since an identity holds for every x , the coefficients of each power of x on both sides must be equal ("equating coefficients").

Worked Example: $(x + a)(x - 5) \equiv x^2 + bx - 15$. Work out a and b .

Expand: $x^2 - 5x + ax - 5a = x^2 + (a - 5)x - 5a$

Constant terms: $-5a = -15 \Rightarrow a = 3$

x terms: $b = a - 5 = -2$

Exam Tip: A single **counter-example** shows that a statement is not an identity. To show $(x + 1)^2 \neq x^2 + 1$ in general, use $x = 2$: $9 \neq 5$.

2.2.5 Algebraic Argument and Proof (A6)

At both tiers you may be asked to use algebra to support or construct an argument, for example to show that a result always holds. The Higher specification adds formal **proof**. Checking a few numerical examples is never enough: the algebra must cover every case.

Standard forms (where n and m are integers):

Even number: $2n$ Odd number: $2n + 1$ Consecutive integers: $n, n + 1, n + 2, \dots$

Consecutive even numbers: $2n, 2n + 2$ Consecutive odd numbers: $2n + 1, 2n + 3$

Multiple of k : $k \times (\text{an integer})$ Any two odd numbers: use different letters, $2m + 1$ and $2n + 1$

Worked Example: Show that the sum of any five consecutive integers is a multiple of 5.

Let the integers be $n, n + 1, n + 2, n + 3, n + 4$

Sum = $5n + 10 = 5(n + 2)$

$n + 2$ is an integer, so the sum is 5 times an integer and therefore a multiple of 5.

Algebraic proof

Higher

Worked Example: Prove that the sum of two consecutive odd numbers is always a multiple of 4.

Let the numbers be $2n + 1$ and $2n + 3$

$$\text{Sum} = 4n + 4 = 4(n + 1)$$

$n + 1$ is an integer, so the sum is a multiple of 4.

Worked Example: Prove that $(n + 4)^2 - n^2$ is a multiple of 8 for every integer n .

$$(n + 4)^2 - n^2 = n^2 + 8n + 16 - n^2 = 8n + 16$$

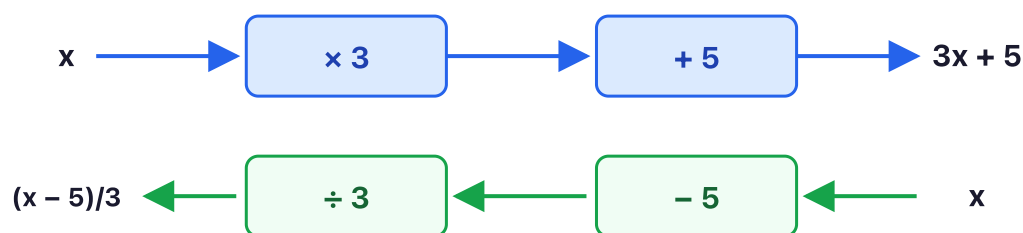
$$= 8(n + 2), \text{ which is 8 times an integer, so it is a multiple of 8.}$$

Exam Tip: Finish every proof with a sentence that links the final expression back to the claim, such as "... = $4(n + 1)$, which is $4 \times$ an integer, so it is a multiple of 4". The final C1 or A1 mark depends on it.

2.2.6 Functions as Inputs and Outputs (A7)

A **function** is a rule that turns each input into exactly one output. Simple expressions can be read as functions, and a function machine sets out the steps in order. To get from an output back to the input, apply the inverse operations in reverse order.

Function machine and its inverse



Worked Example: A function machine multiplies by 3 then adds 5. The output is 38. Work out the input.

Reverse the machine: subtract 5, then divide by 3

$$38 - 5 = 33, \quad 33 \div 3 = 11$$

Input = 11 (check: $11 \times 3 + 5 = 38$ ✓)

2.2.7 Function Notation, Inverse and Composite Functions (A7)

Higher

At Higher tier 1MA1 expects formal function notation, and you need to interpret the reverse process as an **inverse function** and one function followed by another as a **composite function**.

$f(x)$ notation: $f(x) = 5 - 3x$ is the rule; $f(-2)$ is the output when the input is -2 .

Composite function $fg(x)$: means $f(g(x))$; apply g first, then f .

Inverse function $f^{-1}(x)$: reverses f , so $f^{-1}(f(x)) = x$.

Worked Example: $f(x) = 5 - 3x$. Work out $f(-2)$ and solve $f(x) = -10$.

$$f(-2) = 5 - 3(-2) = 5 + 6 = 11$$

$$5 - 3x = -10 \Rightarrow 3x = 15 \Rightarrow x = 5$$

Composite functions

Worked Example: $f(x) = 3x + 1$ and $g(x) = x^2 - 2$. Work out $fg(2)$ and find $gf(x)$.

$$g(2) = 4 - 2 = 2, \text{ so } fg(2) = f(2) = 7$$

$$gf(x) = g(3x + 1) = (3x + 1)^2 - 2$$

$$= 9x^2 + 6x + 1 - 2 = 9x^2 + 6x - 1$$

Exam Tip: $fg(x)$ and $gf(x)$ are usually different. Work from the inside out: in $fg(x)$ the function next to x , which is g , acts first.

Inverse functions

Write $y = f(x)$, rearrange to make x the subject, then replace y with x .

Worked Example: Find $f^{-1}(x)$ when $f(x) = 4x - 9$.

$$y = 4x - 9$$

$$y + 9 = 4x \Rightarrow x = \frac{y + 9}{4}$$

$$f^{-1}(x) = \frac{x + 9}{4}$$

Worked Example: $f(x) = 2x + 5$ and $g(x) = x - 1$. Solve $f^{-1}(x) = g(x)$.

$$f^{-1}(x) = \frac{x - 5}{2}$$

$$\frac{x - 5}{2} = x - 1 \Rightarrow x - 5 = 2x - 2$$

$$x = -3 \text{ (check: } f^{-1}(-3) = -4 \text{ and } g(-3) = -4 \checkmark)$$

Exam Tip: $f^{-1}(x)$ does **not** mean $\frac{1}{f(x)}$. Here the -1 stands for "inverse function", not a reciprocal.

Quick Reference Summary

Skill	Key idea
Using formulae (A5)	Write the formula, substitute, evaluate in the correct order
Changing the subject (A5)	Inverse operations on both sides, in reverse order
Powers and roots (A5)	Isolate the power, then take the root ($\pm$ unless told positive)
Subject appears twice Higher	Collect, factorise, divide
Equation vs identity (A6)	Equation: particular values; identity ($\equiv$): all values
Argument (A6)	Use $2n, 2n + 1, n, n + 1, \dots$ and finish with a conclusion
Proof (A6) Higher	General algebra for every case, plus a concluding statement
Function machines (A7)	Reverse the operations to find an input
$fg(x)$ Higher	$f(g(x))$: g first, then f
$f^{-1}(x)$ Higher	Write $y = f(x)$, make x the subject, swap letters